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RECEIVED 08 December 2025

REVISED 13 February 2026

ACCEPTED 18 February 2026

PUBLISHED 11 March 2026

CITATION

Ironsi CS and Bostanci HB (2026)
Enhancing the reading comprehension
and autonomy of foreign language
students using generative AI-integrated
assistant.

Front. Educ. 11:1763238.

doi: 10.3389/feduc.2026.1763238

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Enhancing the reading comprehension and autonomy of foreign language students using generative AI-integrated assistant

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Generative AI tools are increasingly present in foreign language classrooms, yet empirical research on their structured use for reading comprehension, particularly at the B1 CEFR level, and their relationship to learner autonomy remains limited. Most existing studies focus on writing or unregulated AI use, leaving a gap in understanding how generative AI can be pedagogically integrated into core reading instruction. Addressing this gap, this study examined an eight-week quasi-experimental study implemented in four intact university EFL classes. ChatGPT was embedded as a supplementary scaffold within a three-phase instructional cycle. This intervention consist of a pre-, while-, and post-reading cycle for vocabulary clarification, sentence-level comprehension, inferencing support, and feedback on learner-generated summaries, alongside explicit instruction in critical AI literacy. Using an explanatory sequential mixed-methods design, quantitative data were analyzed with linear mixed-effects models accounting for classroom clustering, with ANCOVA reported as supplementary. Results indicated that participation in the experimental group was associated with higher reading comprehension and perceived autonomy, while qualitative findings showed a shift toward selective, strategic AI use. Teachers can carefully integrate generative artificial intelligence in their classroom to improving students reading comprehension. The study offers a replicable model for teachers to integrate generative AI into B1 reading instruction.

KEYWORDS

B1 CEFR, ChatGPT, generative artificial intelligence, learner autonomy, mixed-methods research, reading comprehension

Introduction

Reading comprehension constitutes a foundational skill in English as a Foreign Language (EFL) learning and serves as a key determinant of academic achievement at the tertiary level. It is conceptualized as a multidimensional construct that encompasses the interaction between lower-level processes, such as lexical access and syntactic parsing, and higher-level processes, including inferencing, integration of ideas, and monitoring of understanding (Istek and Ironsi, 2024). At intermediate proficiency levels, such as CEFR B1, learners are expected to progress beyond literal comprehension by constructing coherent mental representations of texts, identifying main ideas, and drawing simple inferences across clauses and paragraphs (Khan and Mutawa, 2021).

Nevertheless, research consistently indicates that many EFL learners experience difficulty coordinating these processes efficiently, especially when faced with unfamiliar vocabulary,

complex sentence structures, or dense informational texts (Khan and Mutawa, 2021). Consequently, instructional approaches that offer scaffolded support while maintaining learners' engagement with the text are regarded as particularly beneficial for this proficiency range. Previous studies have shown that guided strategy instruction, such as vocabulary previewing, sentence-level clarification, and explicit inferencing practice and use of good lesson planning frameworks, can facilitate reading comprehension development when learners are encouraged to actively process texts rather than depend on external explanations (Istek and Irons, 2024).

Within reading instruction, autonomy is closely associated with the use of metacognitive strategies, including planning approaches to texts, monitoring comprehension breakdowns, and determining when and how to seek support (Al Anqoudi et al., 2023). Evidence indicates that learners who perceive greater autonomy are more likely to engage in strategic reading behaviors and to persist when facing challenges (Khan and Mutawa, 2021). In contrast, excessive dependence on external aids, such as teachers, peers, or technological tools, may diminish opportunities for autonomous regulation if not carefully structured (Al Anqoudi et al., 2023). The integration of digital tools into reading instruction presents a pedagogical tension. While these tools can reduce cognitive load and offer timely assistance, they may also transfer control away from learners if they replace, rather than support, strategic decision-making. Recent scholarship underscores the significance of pedagogical framing in determining whether technologies and digital resources like Artificial intelligence (AI) use enhances or constrains learner autonomy (Xu et al., 2019).

Meanwhile, artificial intelligence (AI) is rapidly reshaping literacy education, transforming how learners access, process, and produce texts in digital environments. Recent systematic reviews show that AI in education now spans adaptive tutoring, automated feedback, and generative tools that support complex literacy practices (Wang et al., 2024; Chen et al., 2024). At the same time, scholars highlight persistent concerns about over-reliance on AI, and the risk of widening existing inequities if such tools are not pedagogically and ethically grounded (Wang et al., 2024; Jiang, 2022). Research admits that in this evolving landscape, generative AI models such as ChatGPT emerge as applicable, dialogic tools that can scaffold reading, explanation, and meaning-making in ways that traditional digital resources cannot easily replicate (Lo et al., 2024; Dempere et al., 2023).

Within language education, research acknowledges that AI already plays a visible role in supporting vocabulary learning, grammar practice, and skills development across proficiency levels (Jiang, 2022; Korkut, 2023). In English as a foreign language (EFL) contexts, research shows that AI-enhanced systems and intelligent tutoring environments have positive effects on reading comprehension by providing adaptive texts, immediate feedback, and individualized practice (Korkut, 2023). Besides, meta-analytic evidence on intelligent tutoring systems indicates moderate positive effects on K–12 students' reading comprehension compared with traditional instruction (Xu et al., 2019).

More recently, a study found that AI-based personalised reading platforms have improved learners' comprehension scores and offered teachers fine-grained diagnostic data (Hidayat, 2024). Adaptivity and continuous feedback are key mechanisms for positive learning impact. Aside from adaptivity and continuous feedback, AI tools designed for reading, such as gamified websites, have also been found to enhance motivation, autonomy, and engagement (Alrabai, 2020). These findings collectively suggest that, when thoughtfully designed, AI can

support both the cognitive and affective dimensions of reading development.

Empirical studies on ChatGPT and related large language models (LLMs) in education are expanding quickly. For instance, a recent systematic review of empirical studies on ChatGPT in ESL/EFL education found that researchers have primarily focused on writing support, feedback generation, and translation, reporting benefits such as increased opportunities for practice, personalization, and reduced anxiety (Lo et al., 2024). However, the same review noted that relatively few studies quantitatively examined the effects of ChatGPT on learners' performance, and that reading, listening, and speaking remain under-researched skills compared with writing (Lo et al., 2024).

Early evidence in the reading domain is promising but partial. In a Turkish EFL context, Çelik et al. (2024) investigated ChatGPT-based simplification of authentic blog texts. They reported significant improvements in university students' reading comprehension and inferential skills, although reading anxiety did not change significantly. Other studies on AI-powered reading platforms that combine tools such as Rewordify, ReadTheory, and ChatGPT have shown that experimental groups using AI-enhanced reading environments can outperform control groups using conventional print or non-adaptive materials on comprehension measures (Hidayat, 2024; Alrabai, 2020). These studies demonstrate that AI-mediated support can make complex texts more accessible and can promote deeper processing. However, they tend to frame AI primarily as a technological innovation, such as a text simplifier or an adaptive platform, rather than as a pedagogically integrated assistant embedded in classroom reading instruction.

Research on AI and learner autonomy further underscores the potential and the complexity of using tools like ChatGPT in literacy development. A recent systematic literature review on artificial intelligence in language learning autonomy found a sharp increase in studies using AI to support autonomous learning. However, it concluded that many of these studies emphasize the functionality of AI applications rather than systematically examining their impact on autonomy, self-regulation, or strategy development (Azhar and Abdullah, 2024a). In related EFL studies, AI-supported reading platforms and self-access tools have been associated with greater learner control, increased motivation, and more independent reading practice (Hidayat, 2024; Alrabai, 2020). However, these investigations rarely articulate explicit pedagogical strategies for transitioning learners from AI dependence to productive, self-regulated use. This issue is particularly salient in lower-intermediate proficiency levels, where learners are still consolidating core reading strategies.

At the B1 level of the Common European Framework of Reference for Languages (CEFR), learners are expected to understand the main points of explicit standard texts on familiar topics, follow straightforward argumentative texts, and locate specific information in everyday materials (Alfiani et al., 2024; Council of Europe, 2020). Designing reading instruction at this level, therefore, involves balancing comprehension support with opportunities to develop inferencing, critical reading, and strategic use of textual clues. Recent studies on CEFR-based digital literacy resources show that aligning digital materials with CEFR descriptors can strengthen task appropriateness and support gradual progression in reading demands (Alfiani et al., 2024). However, there is little empirical

research on how generative AI tools, such as ChatGPT-based assistants, can be integrated into B1 reading lessons to scaffold these goals without undermining learners' active engagement with the text.

Several research gaps emerge from this body of literature. First, although AI and ChatGPT are increasingly used in language education, there is a lack of classroom-based studies examining pedagogical strategies for integrating a ChatGPT-based assistant into reading instruction at a clearly defined proficiency level, such as B1. Existing AI reading studies either focus on generic adaptive platforms (Hidayat, 2024; Alrabai, 2020) or studies using different kinds of research designs (Çelik et al., 2024), rather than on sustained, curriculum-embedded use with explicit teacher mediation.

Second, systematic reviews of ChatGPT in ESL/EFL education highlight a strong emphasis on writing-related applications and call for more rigorous, performance-based research on other language skills, particularly reading (Lo et al., 2024). Third, the autonomy literature notes that research has often privileged technical descriptions of AI tools over nuanced analyses of how AI reshapes learners' sense of control, strategy use, and responsibility for learning (Azhar and Abdullah, 2024a). This is especially significant at the B1 level, where learners are transitioning from heavily teacher-directed reading tasks to more independent engagement with longer and more complex texts.

Addressing these gaps is essential for both theoretical and practical reasons. Theoretically, understanding how a ChatGPT-based assistant can be orchestrated within B1 reading lessons contributes to current debates on the “dual role” of AI in literacy—as both a robust scaffold and a potential source of over-reliance. Empirically, there is a need for mixed-methods studies that not only measure changes in reading comprehension but also capture learners' evolving perceptions of autonomy, their interaction patterns with AI, and their reflections on the benefits and risks of using such tools. Pedagogically, teachers need concrete models for designing prompts, structuring in-class and out-of-class reading activities, and regulating AI support so that learners use the tool to extend, rather than replace, their own comprehension efforts.

The present mixed-methods study responds to these needs by evaluating pedagogical strategies for integrating a ChatGPT-based assistant into B1 CEFR reading instruction. Quantitatively, the study examines the impact of these strategies on learners' reading comprehension, including literal understanding, inferencing, and information integration. Qualitatively, it explores how B1 learners experience ChatGPT as a reading assistant, how they perceive its effects on their autonomy and confidence, and how teachers and students negotiate boundaries between helpful support and dependency. By focusing on a clearly defined proficiency band and embedding ChatGPT use in an authentic instructional sequence, the study moves beyond one-off tool evaluations toward a more situated understanding of AI-mediated literacy practices.

This research makes several contributions. It proposes a pedagogical model for “from text to tool” integration, specifying how teachers can frame ChatGPT as a supplementary assistant rather than a primary source of answers in B1 reading lessons. It offers empirical evidence on the extent to which such integration enhances reading comprehension and supports the development of autonomous reading behaviors. It also contributes to ongoing discussions about ethical and pedagogical safeguards by articulating design principles for responsible AI use in literacy instruction at lower-intermediate levels.

Accordingly, the study addresses the following overarching research question:

- How does integrating a ChatGPT-based assistant through explicit pedagogical strategies into B1 CEFR reading instruction influence students' reading comprehension and perceived learning autonomy compared with traditional reading instruction without AI support?

Sub-questions were generated from the main research question. These include:

- To what extent does a ChatGPT-based reading assistant improve B1 learners' reading comprehension compared with traditional instruction?
- How does the use of the ChatGPT-based assistant influence learners' perceived autonomy and motivation in reading?
- How do students and teachers experience and evaluate the pedagogical and ethical implications of using ChatGPT in B1 reading instruction?

By answering this question, the study aims to inform educators, curriculum designers, and policymakers seeking to leverage generative AI to genuinely enhance literacy learning while maintaining a strong commitment to equity, autonomy, and critical engagement with texts.

Literature review

The rapid expansion of AI-enhanced reading environments provides an important backdrop for understanding how a ChatGPT-based assistant may reshape B1 reading instruction. Early literature on intelligent tutoring systems (ITS) for reading already shows that adaptive, feedback-rich technologies can yield substantial learning benefits. In a meta-analysis of K–12 ITS studies, Xu et al. (2019) reported a moderate overall effect ($g = 0.60$) on reading comprehension, attributing improvements to immediate, individualized feedback and adaptive task sequencing. Subsequent AI-based reading platforms build on this logic by tailoring texts and tasks to learners' profiles, suggesting that personalization and real-time support are central mechanisms through which AI influences reading outcomes.

AI-based personalised reading platforms and reading outcomes

Recent experimental studies confirm that AI-based personalised reading platforms can significantly improve reading comprehension in secondary and EFL contexts. Other authors found that high school students who used an AI-based personalised reading platform significantly outperformed a control group on a standardized reading comprehension test, with the platform recommending texts and providing feedback aligned with each learner's level and interests (Istek and Irons, 2024). Similarly, other authors showed that an AI-based personalised platform and augmented reality significantly enhanced Arab EFL learners' reading comprehension and learning outcomes (Khan and Mutawa,

2021; Tian and Ironsi, 2025). These studies argue that adaptive text selection and comprehension feedback were key when these tools were applied.

These studies converge in demonstrating that personalised AI reading environments can raise comprehension scores and engagement. However, they treat AI primarily as a self-contained platform rather than an interactional assistant embedded in classroom pedagogy. Their outcome measures emphasize test performance and general motivation, while process-oriented constructs such as strategy use, critical engagement with AI suggestions, and perceived autonomy in reading remain under-explored. Moreover, neither study explicitly situates its research design at a clearly defined CEFR level, limiting its transferability to B1-focused curriculum design.

Gamified reading tools, motivation, and autonomy

Parallel studies on gamified reading platforms such as ReadTheory highlight psychological outcomes more explicitly. Drawing on a mixed-methods design, other authors reported that EFL undergraduates who used digital tools perceived improvements in reading skills, motivation, and autonomy; learners emphasized the roles of badges, points, and immediate feedback in sustaining voluntary reading outside class (Tian and Ironsi, 2025). In a more recent study, Al Anqoudi et al. (2023) showed that intensive reading using ReadTheory increased Omani EFL learners' reading motivation and confidence, with students describing the platform as a self-paced environment that encouraged them to seek additional English texts independently.

Taken together, these studies illustrate that AI-supported or AI-adjacent reading tools can foster affective and motivational benefits alongside improvements in comprehension. However, they typically position the digital tool as an autonomous out-of-class resource and seldom analyse how teacher guidance or explicit pedagogical strategies shape learners' evolving autonomy. In addition, they rarely examine how such tools affect the quality of reading processes, which are central to B1 literacy development.

AI-mediated language instruction, motivation, and self-regulated learning

From a psychological perspective, one of the most influential studies is Wei's mixed-method investigation of an AI-mediated language instruction platform in a university EFL course (Wei, 2023). Wei found that students in the AI condition significantly outperformed a control group on measures of English achievement and reported higher L2 motivation and more use of self-regulated learning strategies. Qualitative data indicated that students experienced the AI platform as a personalized, motivating environment that supported planning, monitoring, and reflection. This study demonstrates that AI can enhance both cognitive and self-regulatory dimensions of language learning, aligning with broader evidence that AI systems contribute to motivation and emotional regulation when grounded in psychological theories such as self-determination (Azhar and Abdullah, 2024b).

However, Wei's study treats language achievement holistically. It does not disaggregate effects by skill, leaving open the question of how AI specifically shapes reading comprehension and reading autonomy at particular proficiency levels. Moreover, while it highlights self-regulation, it does not examine how learners negotiate

the boundary between constructive reliance on AI and over-dependence in comprehension tasks—an issue central to the present study.

Generative AI and large language models in reading and assessment

The emergence of large language models (LLMs) such as GPT-3 and ChatGPT extends AI's role from adaptive delivery to the generation of reading materials and tasks. Bezirhan and von Davier used GPT-3 to generate reading passages aligned with existing items for computer-based assessments; human ratings indicated that AI-generated passages could match or exceed original texts in coherence, readability, and appropriateness for young readers (Xiao et al., 2023). Xiao et al. (2023) evaluated reading comprehension exercises automatically generated by ChatGPT for middle-school English learners in China, showing that LLM-generated items were comparable in quality to human-written exercises and enabled high levels of personalization (Irons, 2024).

While these studies demonstrate that generative AI can supply high-quality reading content and exercises, they are primarily technical evaluations. Learners appear mainly as passive recipients or judges of materials, and psychological constructs such as comprehension strategy use, motivation, or autonomy are not foregrounded. Moreover, generative AI is conceptualized as an author of texts rather than as an interactive assistant with whom learners can negotiate meaning, ask questions, or receive formative feedback during reading.

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ChatGPT in language education: current evidence and blind spots

Systematic reviews of ChatGPT in language education underscore both the field's rapid growth and its uneven focus. Ironsi's review of the first year of ChatGPT-language learning research found that most studies relied on small samples, short experimental studies, and self-report data, with limited rigorous evidence on learning outcomes (Irons and Ironsi, 2025). Focusing specifically on ESL/EFL settings, Lo et al. (2024) analysed 70 empirical studies and concluded that ChatGPT has been most used for writing assistance, feedback, and translation; reading received comparatively little attention, and only a minority of studies employed experimental or mixed-method designs. A complementary review of ChatGPT for ESL/EFL writing reported benefits in idea generation, language accuracy, and reduced anxiety, but also highlighted concerns about over-reliance, critical thinking, and academic integrity (Li and Ironsi, 2025).

These reviews converge in identifying a pronounced skill imbalance and methodological limitations. ChatGPT research tends to prioritize writing over reading, examine short-term or hypothetical uses rather than sustained classroom integration, and emphasize perceptions or acceptance rather than performance and process measures. Consequently, there is little empirical guidance on how a ChatGPT-based assistant might be orchestrated pedagogically to support reading comprehension, especially at intermediate proficiency levels.

Empirical studies on ChatGPT and reading

A small but growing body of empirical studies begins to address ChatGPT's role in reading, though often in exploratory or narrowly focused ways. Equally, a provided some strategies that could be applied in integrating GenAI to teach basic English language skills (Irons, 2024). Yousefi investigated the impact of ChatGPT on Iranian university EFL learners' reading comprehension in a quasi-experimental design and found that the experimental group receiving AI support showed substantially larger improvements than the control group (Bezirhan and von Davier, 2023). The study attributed improvements to ChatGPT's ability to provide explanations and additional practice, but it did not systematically examine learners' strategy use, perceptions of autonomy, or potential over-dependence.

Beyond controlled experiments, studies on ChatGPT as a self-learning tool reveal positive but complex psychological dynamics. Tram et al. (2024) conducted a multi-method study of EFL learners who used ChatGPT for self-directed English learning; learners reported high perceived usefulness and convenience, and behavioral data indicated substantial out-of-class engagement (Yousefi and Askari, 2024). However, the study relied primarily on global self-report measures and focused on general language learning, making it difficult to infer how ChatGPT specifically supports reading comprehension or equips learners with transferable reading strategies.

Collectively, these studies suggest that ChatGPT can enhance reading outcomes and support personalized learning. However, they leave key questions unanswered: under what instructional conditions do such benefits arise, how do learners cognitively and affectively engage with ChatGPT during reading, and how does AI support intersect with learners' sense of autonomy and responsibility?

AI and language learning autonomy

Research on AI and language learning autonomy provides another important strand. Azhar and Abdullah's systematic review of AI in Language Learning Autonomy (AILLA) found that AI-supported tools often promote self-paced learning and independent practice. However, most studies focus on describing the workings of AI applications rather than empirically analysing changes in autonomy-related behaviors or beliefs (Tram et al., 2024). The review noted a predominance of mobile and chatbot applications, with limited attention to generative AI and to integrating autonomy frameworks from educational psychology (such as self-determination theory or self-regulated learning models).

This pattern mirrors broader trends in the AI-language learning literature: while motivation, engagement, and self-regulation are mentioned, they are seldom operationalized alongside fine-grained measures of skill development within specific tasks such as reading at the B1 level. Thus, the psychological mechanisms through which AI—especially ChatGPT—shapes learners' autonomy in reading remain theoretically underdeveloped and empirically under-tested.

Synthesis and gaps

Across these strands, several converging insights and gaps emerge. AI-based personalised and gamified reading platforms consistently improve comprehension and enhance motivation, but they rarely address how learners critically engage with AI support or develop autonomous reading strategies (Istek and Irons, 2024; Khan and

Mutawa, 2021; Tian and Irons, 2025; Al Anquoudi et al., 2023). Generative AI research demonstrates that LLMs can create high-quality reading texts and exercises. However, it essentially treats learners as recipients of AI-generated content rather than as dialogic partners in meaning-making (Irons, 2024; Xiao et al., 2023). ChatGPT-focused studies and reviews highlight positive perceptions and potential learning benefits. However, they emphasize writing, rely heavily on self-report, and offer limited evidence about sustained, classroom-based reading instruction at clearly defined proficiency levels (Irons and Irons, 2025; Lo et al., 2024; Li and Irons, 2025; Bezirhan and von Davier, 2023; Yousefi and Askari, 2024). Autonomy-oriented reviews highlight AI's promise for self-directed learning while lamenting the lack of theoretically grounded, empirical studies on how AI reshapes autonomy in practice (Tram et al., 2024).

The present study is designed to address these gaps by examining a ChatGPT-based assistant not as a standalone platform or content generator but as a pedagogically orchestrated tool embedded in B1 CEFR reading instruction. It responds to the documented need for (a) rigorous, mixed-method research on ChatGPT's impact beyond writing, (b) skill-specific analyses of AI-supported learning at intermediate proficiency levels, and (c) theoretically informed investigations of how AI-mediated scaffolding influences learners' reading comprehension and perceived autonomy. By doing so, the study aims to move the conversation beyond technological feasibility toward a psychologically and pedagogically grounded understanding of how generative AI can be harnessed to support, rather than supplant, learners' active engagement with texts.

The theoretical and conceptual framework for this study integrates contemporary models of reading, sociocultural perspectives on scaffolding, and motivational theories of autonomy and self-regulation. Together, these frameworks explain how a ChatGPT-based assistant embedded in B1 reading pedagogy can influence learners' reading comprehension and perceived autonomy. From a cognitive standpoint, recent interactive and dynamic models of reading conceptualise comprehension as the result of interactions among word reading, language skills (e.g., vocabulary, inference), background knowledge, and higher-order cognitive processes (Zhang and Liu, 2024).

These models emphasise that readers actively construct meaning by integrating textual information with prior knowledge and by deploying strategies such as inferencing and monitoring. Empirical applications of interactive reading models in EFL and L1 contexts show that instruction that explicitly targets these processes leads to improvements in comprehension and strategy use (Kim, 2020). In the present study, this perspective underpins the design of reading tasks and the focus on B1-level comprehension processes (literal understanding, inference, and information integration).

The sociocultural tradition further highlights that tools and social interaction mediate reading development. Scaffolding—temporary support that enables learners to accomplish tasks they could not yet complete independently—plays a central role in helping EFL readers construct meaning from texts (Ariani, 2023; Yawiloeng, 2021). Recent studies demonstrate that both peer and teacher scaffolding enhance EFL learners' reading comprehension and vocabulary, with different patterns of support affecting how learners engage with texts and discourse (Dehghan, 2013; Abdulaal, 2024). Within this view, the ChatGPT-based assistant is conceptualised as a mediating artefact that provides contingent, dialogic support (e.g., explanations, questions, feedback) while teachers orchestrate when and how the tool is used. The framework assumes that effective integration requires deliberate

teacher mediation and explicit instruction in “critical AI literacy” so that scaffolding gradually shifts from AI- and teacher-led to learner-led control.

Motivationally, the study draws on Self-Determination Theory (SDT) and research on self-regulated learning (SRL). SDT posits that learning environments that support autonomy, competence, and relatedness foster more self-determined forms of motivation and sustained engagement (Mojarrabi, 2024; McEown, 2019; Mynard, 2022). Recent SDT-informed studies in language education show that when support systems are designed to address these needs, learners exhibit higher L2 achievement, greater persistence, and greater wellbeing (Mojarrabi, 2024; Al-Hoorie et al., 2022). SRL theory complements this perspective by describing how successful learners plan, monitor, and regulate their cognition, motivation, and behavior during learning tasks (Ma et al., 2025; Zimmerman and Schunk, 2011). AI-mediated environments have been shown to promote self-regulatory behaviors when they provide opportunities for goal setting, monitoring, and reflection (Wei, 2023). In this study, the ChatGPT-based assistant is theorized to support autonomy (choice of questions, control over the timing of help), competence (tailored feedback, accessible explanations), and SRL (prompting strategy use and reflection), thereby increasing perceived reading autonomy.

Conceptually, the framework positions generative AI not as a replacement for human instruction but as an additional layer of scaffolding within an interactive, need-supportive learning environment. It provides a coherent basis for the study’s hypotheses and mixed-method design by specifying the mechanisms (cognitive, social, and motivational) through which ChatGPT integration is expected to affect reading outcomes at the B1 level.

Figure 1 (conceptual framework) synthesizes these strands. At the left, experimental group B1 reading pedagogy (task design, prompts, teacher mediation, and critical AI literacy principles) shapes the nature of AI-mediated scaffolding and interaction patterns (pre-/while-/post-reading support, adaptive feedback, student–AI dialogue). These, in turn, influence learner processes, comprising cognitive–strategic processes (inferencing, monitoring, and vocabulary use) and motivational–autonomy processes (perceived autonomy,

competence, SRL strategies). These learner processes and interaction patterns jointly predict the primary outcomes: B1-level reading comprehension and perceived reading autonomy. The framework also acknowledges moderating factors such as initial proficiency, prior AI experience, and baseline motivation, which may condition the magnitude of the effects.

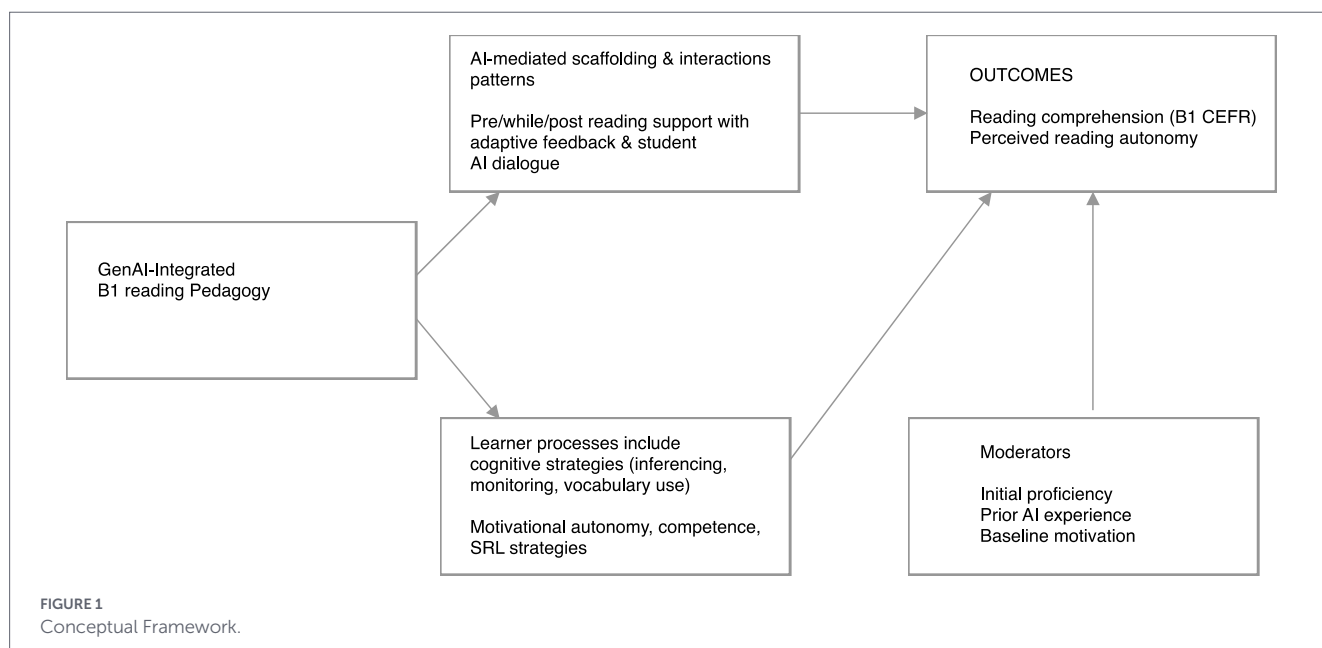
Methods

Research design

This study adopted an explanatory sequential mixed-methods design, combining quantitative and qualitative approaches (Zimmerman, 2002). The first phase comprised a quasi-experimental pretest–posttest control-group design to examine the effects of integrating a ChatGPT-based assistant into B1 reading instruction on students’ reading comprehension and perceived autonomy. The second phase used qualitative methods to explain and elaborate on the quantitative patterns. Integration occurred at the design, methods, and interpretation levels, with joint displays used to connect quantitative results and qualitative themes, as required for this research design (Creswell and Plano Clark, 2018). This design was appropriate because it enabled the study to address the primary causal-comparative research question while capturing learners’ and teachers’ experiences of AI-mediated reading.

Context, participants, and sampling

The study took place in a B1 English reading course at a private university. Four intact class groups (two morning and two afternoon sections) participated. Intact classes were chosen to preserve naturally occurring instructional settings and minimise disruption. Using cluster allocation, two classes ($n = 44$) were assigned to the experimental group and two classes ($n = 42$) to the control group (traditional instruction). A total of 86 students participated in the study.



Participants were aged 18–22 years ($M = 19.4$) and had diverse L1 backgrounds; all had achieved B1 on the institution's CEFR-aligned placement test.

A convenience sampling strategy was employed because access to classes was constrained by institutional scheduling, a common approach in quasi-experimental pretest–posttest research design (Zimmerman, 2002). Power analysis for repeated-measures ANCOVA ($\alpha = 0.05$, power = 0.80, medium effect size) indicated a minimum of 70 participants; the final sample exceeded this threshold, allowing for attrition (Fetters et al., 2013; Field, 2018). Participants for the qualitative phase were selected using purposeful maximum-variation sampling based on quantitative learning gains. Gain scores were calculated separately for reading comprehension and perceived reading autonomy by subtracting pretest scores from posttest scores. Students were categorized as High gain (top 33% of gain scores), Medium gain (middle 33%), and Low gain (bottom 33%).

From the experimental classes, 24 students were selected to ensure representation across gain levels (8 high, 8 medium, 8 low). This approach allowed examination of how students with differing quantitative outcomes experienced and engaged with the ChatGPT-integrated pedagogy. A total of four focus groups were conducted with student participants. These groups consist of 5–7 students per group, 3 focus groups from the experimental condition, and 1 focus group from the control condition (used for contrastive purposes).

Experimental focus groups were composed to include a mix of high-, medium-, and low-gain students. The control focus group included students with comparable proficiency levels but did not use ChatGPT. All focus groups were semi-structured, lasted approximately 45–60 min, and were conducted in English with occasional L1 support when necessary to ensure clarity of expression.

Two experimental and two control lessons were observed during weeks 3 and 7. Non-participant classroom observations were conducted in Weeks 3 and 7 of the study to capture early-stage and later-stage instructional dynamics. Observations focused on students interacting with texts, consultation with ChatGPT, teacher guidance, and regulation of AI use, as well as evidence of independent strategy use and verification behavior. Field notes were recorded using a structured observation guide. Observation data were not analyzed as a standalone dataset, but were used to corroborate, refine, and contextualize themes emerging from focus group and interview data.

Class assignment and teacher distribution

The study was conducted with four intact B1-level university reading classes offered during the same academic term. Due to institutional scheduling constraints, assigning students randomly was not feasible. Instead, intact classes were assigned to instructional conditions at the class level, yielding two experimental and two control classes.

Each class was taught by one of four instructors. Each teacher taught only one class, and therefore, teachers were not crossed or balanced across instructional conditions. Two teachers taught the experimental classes, and two different teachers taught the control classes. As a result, teacher and class membership were fully confounded, such that each class had a unique teacher.

Assignment of classes to condition occurred prior to the start of the study and was based on timetable availability rather than student characteristics or prior achievement. All classes followed the same institutional curriculum and were operating at the same CEFR B1

proficiency level as determined by university placement procedures. To reduce instructional variability across classes, several standardization measures were implemented prior to and during the study implementation.

First, all classes followed a shared syllabus that specified weekly instructional objectives, pacing, and learning outcomes aligned with the CEFR B1 reading descriptors. Second, all teachers used the same set of core reading texts from the department's official course materials. These texts were selected in advance and used consistently across all four classes. Third, common assessments were employed across conditions. The same reading comprehension pretest and posttest, as well as the same perceived autonomy questionnaire, were administered to all students following identical procedures and timelines. Teachers did not have access to test items prior to administration.

Finally, before commencing the study, all four instructors participated in a brief coordination meeting led by the research team. During this meeting, the overall study goals, instructional timeline, and assessment procedures were explained. Teachers assigned to the experimental condition received additional guidance on the pedagogical principles underlying the ChatGPT-integrated reading activities, including rules for AI use, emphasis on verification against the source text, and the framing of ChatGPT as a supplementary scaffold rather than a replacement for reading. Teachers in the control condition were instructed to follow the standard reading pedagogy outlined in the shared syllabus, without using generative AI tools. Although these procedures aimed to promote instructional consistency, the study did not include formal monitoring or calibration of teacher instructional style, and teachers retained autonomy in their day-to-day classroom interactions.

Instruments

A researcher-developed test was used to collect data from the participants. This test aligned with B1 CEFR descriptors and included four texts (emails and short expository and opinion texts) and 30 items targeting literal, inferential, and integrative comprehension. At this level, readers are expected to understand the main points of explicit, standard texts on familiar topics; locate specific information; make straightforward inferences; and integrate information across sentences and paragraphs (Council of Europe, 2020). To operationalize this construct, the test targeted four interrelated reading skills commonly associated with B1 comprehension: literal comprehension, inferential comprehension, integrative comprehension, and summarization of main ideas. The test was developed to reflect both process-oriented and outcome-oriented aspects of reading comprehension, consistent with CEFR descriptors and prior EFL reading assessment research.

The reading comprehension test consisted of four texts and 30 items. Table 1 presents the test blueprint, including the number of items per reading skill, text genres, and alignment with CEFR B1 descriptors.

The four reading texts represented common academic and general-interest genres appropriate for university-level B1 learners. These include: informational articles, short expository texts, narrative-based informational texts, and opinion-neutral explanatory passages. All texts were between 450 and 550 words in length and were screened to ensure lexical and syntactic complexity consistent with B1-level expectations.

TABLE 1 Reading comprehension test blueprint.

Reading skill	Number of items	Item format	CEFR B1 alignment
Literal comprehension	10	Multiple choice	Identifying explicitly stated information; understanding factual details
Inferential comprehension	8	Multiple choice	Making simple inferences; understanding implicit meaning
Integrative comprehension	6	Multiple choice	Integrating information across sentences/paragraphs; identifying main ideas.
Summarization	6	Short written response	Producing a summary that captures the main points of a text
Total	30	—	—

The reading comprehension test employed a parallel forms design, with separate pretest and posttest versions. The two forms were constructed to be comparable in structure and difficulty, following the exact test blueprint. Specifically, both forms contained four texts of similar length (± 30 words). The texts were matched for genre distribution. Lexical profiles were examined using frequency-based word lists to ensure comparable vocabulary load. Each form included the same number of items per reading skill and identical item formats. The two versions were developed concurrently and piloted with a comparable group of B1-level students prior to the main study. Internal consistency reliability for both forms exceeded acceptable thresholds ($KR-20 > 0.80$). However, formal statistical equating procedures were not conducted, which represents a limitation. As a result, although the two forms were designed to be parallel, perfect equivalence cannot be assumed.

Summarization responses were scored using an analytic rubric focusing on three dimensions. First was based on content coverage (idea units). This includes the main ideas presented in the text. Second was based on accuracy and relevance, and this pertained to how the writer used accurate information as the source text and avoided irrelevant or incorrect information. Third was based on coherence and clarity, especially logical organization and clarity of expression. Each dimension was rated on a 0–2 scale, yielding a maximum of 6 points per summary. Summarization scores were then combined with multiple-choice scores to produce the total reading comprehension score. All summarization responses were anonymized prior to scoring.

For inter-rater reliability, summarization responses were scored by two trained raters, both with graduate-level training in applied linguistics and experience teaching EFL reading. Prior to scoring, the raters participated in a calibration session during which the rubric was discussed, and sample responses were jointly evaluated. To assess scoring reliability, 25% of the summaries were independently scored by both raters. Inter-rater reliability was estimated using the intraclass correlation coefficient (ICC, two-way random effects, absolute agreement). The resulting ICC was 0.87, indicating high inter-rater reliability. The remaining responses were scored by one rater.

Perceived reading autonomy was measured using an adapted self-report questionnaire drawing primarily on established instruments grounded in self-determination theory and learner autonomy research. The core items were adapted from the Learner Autonomy Questionnaire (Spratt et al., 2002), the Learner Autonomy Scale for EFL contexts (Oğuz, 2013), and the Academic Autonomy Scale (Macaskill and Taylor, 2010). These instruments have been widely used to assess learners' perceived responsibility, decision-making, and self-regulation in language learning contexts. From these sources, 24 items were selected and adapted to focus specifically on reading-related autonomy in an EFL university context. All items were rated

on a 5-point Likert scale ranging from 1 (*strongly disagree*) to 5 (*strongly agree*), with higher scores indicating greater perceived reading autonomy.

Prior to the main study, an exploratory factor analysis (EFA) was conducted using pilot data ($n = 32$) to examine the internal structure of the adapted scale. Given the small sample size relative to the number of items, this analysis was treated as exploratory and tentative. It was used solely to identify potentially problematic items rather than to establish a definitive factor structure. Although a three-factor solution emerged in the pilot analysis, no strong claims are made regarding the factorial validity of the scale. Accordingly, the present study focuses primarily on total autonomy scores, and any interpretations of subscales were treated with caution.

Internal consistency reliability was examined using the main study sample ($N = 86$). Reliability coefficients were calculated for the total autonomy scale and for each tentative subscale. For the Total autonomy scale, Cronbach's $\alpha = 0.91$ was obtained, while $\alpha = 0.88$ was obtained for the Strategy use and self-regulation subscales. Cronbach's alpha scores of $\alpha = 0.84$ and $\alpha = 0.81$ were obtained for the subscales Decision-making and responsibility and Self-evaluation and reflection. These values indicate good to excellent internal consistency, supporting the use of the scale as a measure of perceived reading autonomy in the present sample. Semi-structured focus-group and interview guides addressed experiences of AI-mediated reading, perceived benefits and challenges, and views on autonomy—an observation protocol captured patterns of teacher–student–AI interaction and strategy instruction.

Procedures

Data collection spanned 10 teaching weeks. In Week 1, the researcher briefed students on the study and obtained informed consent. All participants completed the reading pretest and autonomy questionnaire during regular class time. All classes worked with the same reading texts and followed a pre-reading, while-reading, and post-reading structure.

In the experimental condition, ChatGPT was used as a supplementary support tool during reading activities. In the pre-reading phase, students previewed key vocabulary and were allowed to use ChatGPT to clarify word meanings after attempting to infer meanings independently. During the while-reading phase, students first read the text individually. Then they used ChatGPT selectively to clarify complex sentences or to support inferencing, with explicit instructions to verify AI responses against the original text. In the post-reading phase, students wrote their own summaries and used ChatGPT only to check coverage of main ideas and identify missing information; generating summaries using AI was not permitted.

Throughout the study, teachers emphasized critical AI literacy, including the need to question AI output, compare it with the source text, and avoid copying AI-generated language. Over time, teacher guidance was gradually reduced to encourage more autonomous reading behavior. From Weeks 2–9, both groups followed the same syllabus, text types, and assessment schedule. In the experimental group, the teacher integrated the ChatGPT-based assistant into the pre-reading stages and post-reading stages in line with the pedagogical model described earlier. Learners used ChatGPT for vocabulary previews, clarification questions, and feedback on short written responses, under explicit guidelines for critical and responsible use.

During Weeks 3 and 7, the researcher conducted non-participant observations in one lesson per class (four lessons in total), taking field notes focused on scaffolding and interaction patterns. In Week 10, all students completed the posttest and post-questionnaire. The control condition followed the same instructional structure, texts, and time-on-task, but did not involve AI tools. Vocabulary explanations, sentence clarification, inferencing, and feedback on summaries were provided through teacher instruction and peer discussion. Instructional time was held constant across groups. Subsequently, focus groups with students and interviews with teachers were conducted in a quiet room on campus, each lasting 45–60 min and audio-recorded.

Data analysis

Quantitative analyses were conducted to examine differences between the experimental and controlled conditions in reading comprehension and perceived reading autonomy. All analyses were performed using an alpha level of 0.05. Quantitative data were analysed using SPSS 29. Following screening for missing data and normality, descriptive statistics and reliability indices (Cronbach's α) were calculated. Data were initially examined using ANCOVA. To account for the nested classroom structure, linear mixed-effects models with random class intercepts were used as the primary analytic approach.

Given the nature of the study, students were nested within intact classes. To account for this hierarchical structure, linear mixed-effects models (LMMs) were used as the primary analytic approach. For each outcome variable, posttest scores served as the dependent variable, instructional condition (experimental vs. control) was entered as a fixed effect, and the corresponding pretest score was included as a covariate. A random intercept for class was specified to account for between-class variability. Fixed-effect estimates are reported as unstandardized coefficients (β), along with standard errors, 95% confidence intervals, and p values.

Because the study involved only four classes, the statistical power to estimate random effects was limited, and the variance component

estimates should be interpreted cautiously. Nevertheless, mixed-effects modeling was selected as the most appropriate approach for addressing the non-independence of observations inherent in the design. For transparency and comparability with prior classroom-based research, ANCOVA analyses were also conducted using posttest scores as the dependent variable, instructional condition as the independent variable, and pretest scores as covariates.

However, given that ANCOVA assumes independent observations and does not account for clustering, these analyses were treated as confirmatory rather than primary and are reported in the [Supplementary materials](#). Assumptions for the mixed-effects models were evaluated by inspecting residual distributions and fitted values. No severe deviations from normality or influential outliers were observed. Descriptive statistics were computed for all variables prior to inferential analysis.

Qualitative data were analysed using reflexive thematic analysis. The thematic analysis adhered to Braun and Clarke's six-phase procedure that includes familiarization, coding, theme development, review, definition, and narrative reporting (Tabachnick and Fidell, 2019; Braun and Clarke, 2022; Byrne, 2022). An abductive stance combined deductive codes derived from the conceptual framework (scaffolding, strategy use, autonomy) with inductive codes emerging from participants' accounts. The coding framework is presented in [Table 2](#).

Coding was conducted by the first author and another independent coder, with regular peer debriefing sessions used to challenge interpretations and enhance reflexivity. Initially, interview recordings were transcribed and examined to gain familiarity with the data. The procedure was repeated several times to ensure that both coders were adequately familiar with the transcript before initiating coding. The coders manually coded the transcript by identifying patterns that reflect an idea.

Initial codes were derived from the data, and codebooks documented the codes that emerged. The coders carefully reviewed the codes, leading to the removal of some, as agreed. The codes were reviewed to ensure only those that aligned with the research questions and study objectives were retained. After reaching a consensus, the codes were reviewed again by an invited coder. Member checking was conducted during this process. These were done to ensure inter-coder reliability, and the coder assisted in resolving conflicts regarding three codes. After this, codes were organized into overarching themes that addressed the research questions. The interviewees were given the themes to ensure that they reflected their opinions. Afterward, the themes were defined, named, and presented in tables.

Observation data were used to refine the theme and for temporal comparison. Observational evidence was used to confirm whether reported behaviors were visible in classroom practice. Differences between Week 3 and Week 7 observations informed interpretations of

TABLE 2 Coding framework.

Code category	Example codes	Source
Strategic reading behavior	Vocabulary inference, rereading, inferencing	Deductive
AI as a scaffold	Clarification support, feedback tool	Deductive
Autonomy development	Independent problem-solving, goal setting	Deductive
Critical AI awareness	Distrust of AI output, verification behavior	Deductive
Behavioral change over time	Reduced AI reliance, selective use	Inductive
Affective responses	Confidence, frustration, reassurance	Inductive

developmental change, particularly in students' strategic use of AI and willingness to attempt comprehension independently. Illustrative observational insights were integrated into the Findings section alongside focus group excerpts to strengthen credibility as warranted by assertions in qualitative research (Creswell and Plano Clark, 2018).

Ethical considerations

The study received approval from the university ethics committee. Participation was voluntary; students could withdraw at any point without academic penalty. Data were anonymized using codes, and all digital files were stored on password-protected drives. Use of the ChatGPT-based assistant complied with institutional data-protection guidelines, and no personally identifiable information was entered into the system. Participants were explicitly reminded that the AI could provide inaccurate information and were encouraged to verify AI outputs against the text and classroom discussion.

Findings

The findings are organized by outcome variable. Linear mixed-effects models accounting for classroom-level clustering are reported as the primary analyses. ANCOVA results are provided in the [Supplementary materials](#) for transparency.

Table 3 summarizes pre- and posttest scores for reading comprehension (0–30 points) and perceived reading autonomy (1–5 scale). At pretest, the two groups show comparable reading scores

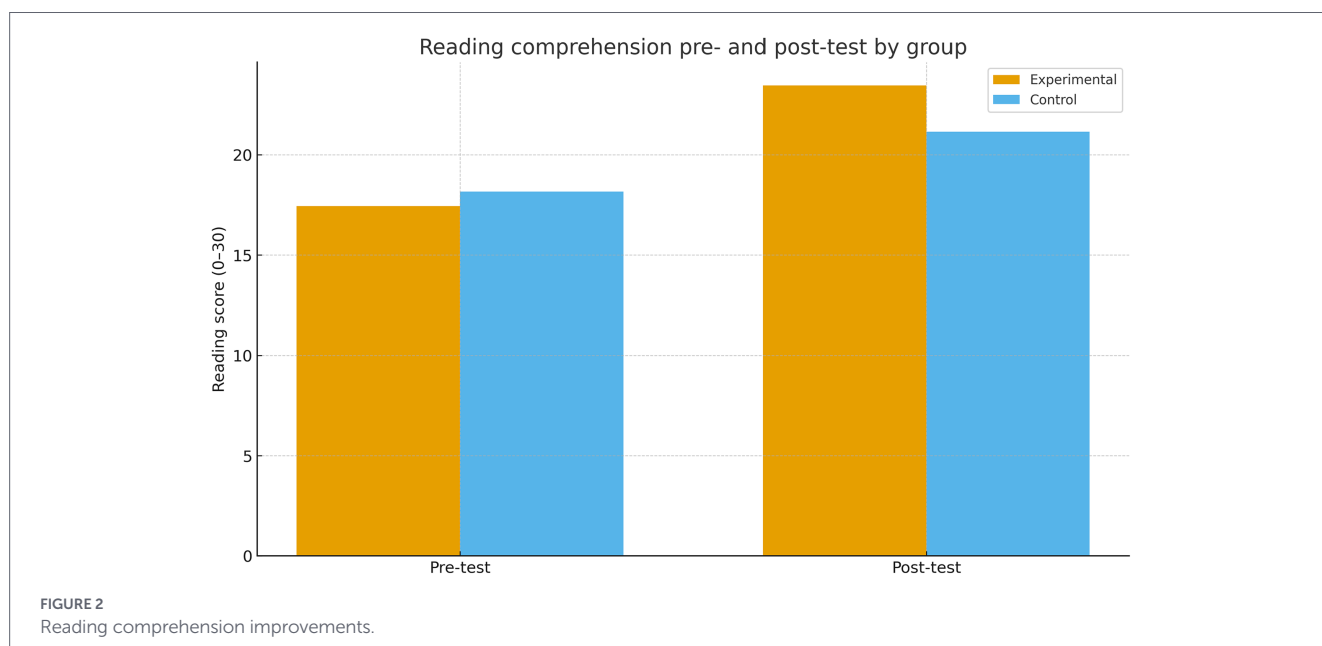
(experimental $M = 17.44$, control $M = 18.17$) and autonomy levels (experimental $M = 3.10$, control $M = 2.99$), indicating similar starting points. After the study, both groups improved in reading comprehension, but the experimental group improved more (mean increase $\approx +6.0$ points) than the control group ($\approx +3.0$ points). A similar pattern emerges for perceived autonomy, with a larger increase in the experimental group ($+0.57$) than in the control group ($+0.25$). Figure 2 depicts improvements in reading comprehension, while Figure 3 illustrates trajectories of autonomy.

Table 4 reports the results of a linear mixed-effects model examining the association between instructional condition and posttest reading comprehension while controlling for students' pretest reading performance and accounting for the nested classroom structure. The fixed effect for instructional condition was positive and statistically significant ($\beta = 3.13$, $SE = 0.39$, 95% CI [2.36, 3.90], $p < 0.001$). This coefficient indicates that, after adjusting for baseline reading ability, students in the experimental group scored on average approximately 3.1 points higher on the posttest reading assessment than students in the control condition. The confidence interval does not include zero, suggesting that this difference is unlikely to be due solely to sampling error.

The fixed effect for pretest reading comprehension was also statistically significant and large in magnitude ($\beta = 1.14$, $SE = 0.07$, 95% CI [1.00, 1.28], $p < 0.001$). This finding indicates that higher baseline reading ability strongly predicted posttest performance. Specifically, for each one-point increase in pretest reading score, posttest reading scores increased by approximately 1.14 points, holding instructional condition constant. This result suggests substantial rank-order stability in reading comprehension over the instructional period. The intercept ($\beta = 0.43$, $p = 0.74$) was not statistically significant. Because

TABLE 3 Descriptive statistics for reading comprehension and perceived autonomy by group.

Group	<i>n</i>	Pretest reading M (SD)	Posttest reading M (SD)	Pretest autonomy M (SD)	Posttest autonomy M (SD)
Experimental	44	17.44 (2.76)	23.45 (3.51)	3.10 (0.40)	3.67 (0.50)
Control	42	18.17 (2.93)	21.14 (3.91)	2.99 (0.43)	3.24 (0.54)



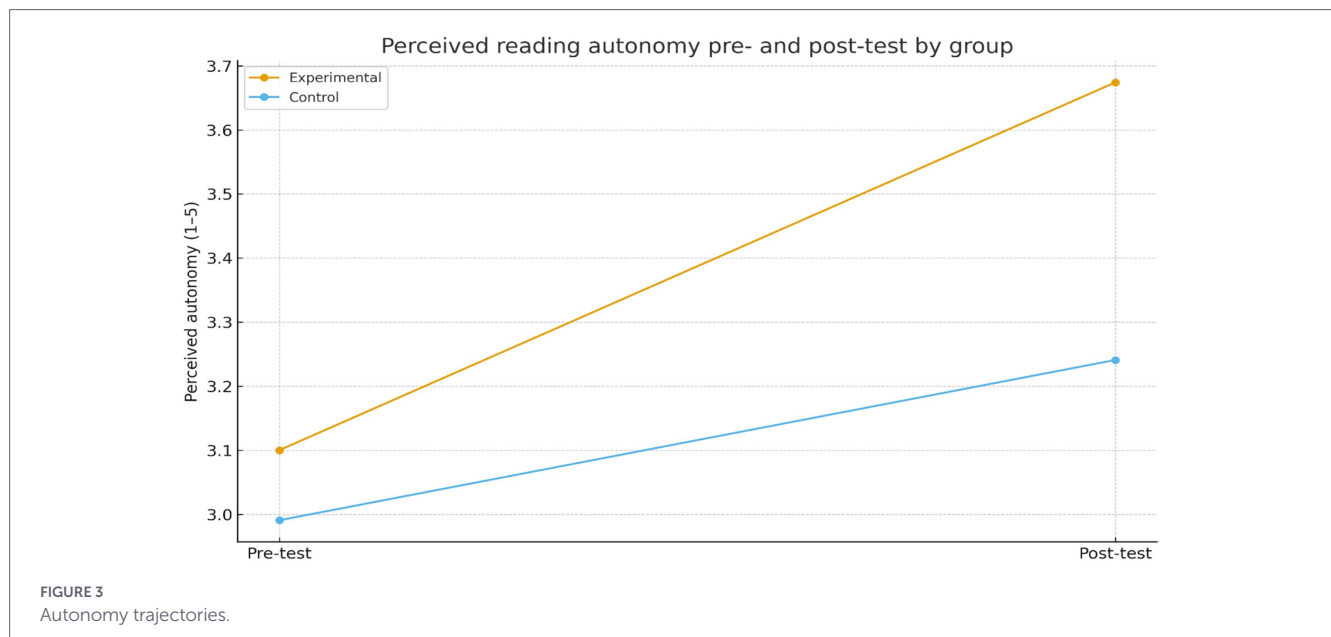


TABLE 4 Linear mixed-effects model predicting posttest reading comprehension.

Fixed effects	Estimate (β)	SE	95% CI	<i>p</i>
Intercept	0.43	1.29	[-2.09, 2.96]	0.74
Condition (experimental)	3.13	0.39	[2.36, 3.90]	< 0.001
Pretest reading	1.14	0.07	[1.00, 1.28]	< 0.001

Random effects.	Variance
Class (intercept)	< 0.001
Residual	—

Condition was coded as 0 = control, 1 = experimental. Pretest reading scores were grand-mean centered. The model includes a random intercept for class. Estimates are unstandardized. CI, confidence interval.

pretest scores were grand-mean centered, the intercept represents the expected posttest reading score for a control-group student with an average pretest score. The non-significant intercept has no substantive interpretive implications for the research questions.

The variance associated with the random class intercept was minimal (approaching zero), indicating limited between-class variability in posttest reading scores after accounting for instructional condition and pretest performance. This suggests that most of the variability in reading outcomes was attributable to individual-level differences rather than class-level differences in this sample. However, given the small number of classes (four), the precision of this variance estimate is limited, and it should be interpreted cautiously. Table 2 indicates that participation in the experimental group was associated with higher posttest reading comprehension, even after controlling for baseline reading ability and accounting for classroom clustering. While the effect size appears meaningful, the quasi-experimental design and limited number of clusters preclude strong causal claims. ANCOVA results yielded a comparable pattern of findings and are reported in Supplementary Table S1.

Table 5 presents the results of a parallel linear mixed-effects model examining differences in perceived reading autonomy between instructional conditions, controlling for baseline

autonomy and accounting for class-level clustering. The instructional condition had a positive, statistically significant effect on posttest autonomy ($\beta = 0.31$, SE = 0.07, 95% CI [0.18, 0.45], $p < 0.001$). This coefficient indicates that, after controlling for autonomy, students in the experimental group reported autonomy scores approximately 0.31 points higher than those in the control condition on the posttest autonomy scale. The relatively narrow confidence interval suggests a stable estimate of this association.

The fixed effect of pretest autonomy was also statistically significant and substantial in magnitude ($\beta = 1.08$, SE = 0.07, 95% CI [0.94, 1.22], $p < 0.001$). This finding indicates that students' initial perceptions of autonomy strongly predicted their post-intervention autonomy. In practical terms, students who entered the course with higher levels of perceived autonomy were likely to maintain or further develop these perceptions over time. The intercept was close to zero and non-significant ($\beta = 0.01$, $p = 0.97$). As with the reading model, the intercept represents the expected posttest autonomy score for a control-group student with average baseline autonomy and is not substantively meaningful for interpretation.

The estimated variance of the random class intercept was slight (Var ≈ 0.02), indicating limited between-class variability in

TABLE 5 Linear mixed-effects model predicting posttest reading autonomy.

Fixed effects	Estimate (β)	SE	95% CI	p
Intercept	0.01	0.22	[−0.41, 0.43]	0.97
Condition (experimental)	0.31	0.07	[0.18, 0.45]	< 0.001
Pretest autonomy	1.08	0.07	[0.94, 1.22]	< 0.001

Random effects	Variance
Class (intercept)	0.02
Residual	—

Condition was coded as 0 = control, 1 = experimental. Pretest autonomy scores were grand-mean centered. The model includes a random intercept for class. Estimates are unstandardized. CI, confidence interval.

TABLE 6 Themes, codes, and subcodes from qualitative analysis.

Theme	Codes	Subcodes
ChatGPT as a flexible scaffold for comprehension	Vocabulary support; clarifying complex parts; checking understanding; practice questions	On-demand glosses; paraphrasing sentences; “second explanation” after teacher; using AI questions as self-tests
From dependence to strategic autonomy	Initial over-reliance; evolving self-regulation; choosing when not to ask; combining AI and personal strategies	“Copy–paste temptation”; self-check before consulting AI; using AI to confirm predictions; delaying help to attempt inference first.
Reconfigured classroom roles and critical awareness of AI	Teacher as coach; new participation patterns; noticing AI errors; discussing reliability and ethics	Teacher-guided prompts; peer collaboration around AI responses; spotting hallucinations; concerns about over-use and fairness

autonomy outcomes after accounting for instructional condition and baseline autonomy. As with the reading model, this suggests that most variability in autonomy perceptions occurred at the individual level. However, the small number of classes limits the stability of this estimate. The results in Table 3 suggest that participation in the experimental group was associated with higher perceived reading autonomy, beyond what would be expected based on students’ initial autonomy levels. These findings align with the qualitative evidence indicating that students increasingly engaged in strategic and self-directed reading behaviors over the course of the study. Confirmatory ANCOVA results were reported in Supplementary Table S2.

Table 6 presents the themes, codes, and subcodes from qualitative analysis. Qualitative analysis addressed RQ3, exploring how students and teachers experienced the ChatGPT-based assistant and how they perceived its pedagogical and ethical implications. Reflexive thematic analysis generated three overarching themes (Table 5).

Theme 1: ChatGPT as a flexible scaffold for comprehension

Students consistently described ChatGPT as an accessible “extra helper” that supported them at different stages of reading.

“When I do not understand a sentence, I paste it to ChatGPT and ask it to explain ‘like I am B1’. Then I can go back to the text and follow the idea.” (S13)

“The vocabulary list from the AI before reading makes the text less scary. I feel I know the topic already.” (S04)

“Sometimes I answer the teacher’s questions and then check my answer with ChatGPT. If we agree, I feel more confident.” (S27)

These excerpts illustrate how learners used the assistant to obtain simplified explanations, pre-teach key vocabulary, and verify their own comprehension. Teachers similarly viewed the tool as a flexible scaffold:

“It is like having another teaching assistant who is always free to answer ‘why’ questions that I do not have time to answer individually.” (T2)

The theme, therefore, captures ChatGPT’s role as a mediating artefact that extends, rather than replaces, teacher scaffolding by providing individually tailored support during pre-, while-, and post-reading phases.

Theme 2: from dependence to strategic autonomy

While students initially consulted ChatGPT frequently, many reported developing a more selective, strategic approach over time.

“At the beginning, I asked ChatGPT for almost every word. Later, the teacher told us to try to guess from context first, so now I only ask when the word is significant.” (S09)

“Sometimes I write my own summary first and then ask ChatGPT to write one. If mine is similar, I feel I am improving.” (S31)

“I realised if I only copy the answer from ChatGPT, I do not remember it. Now I use it more to check my ideas, not to produce them.” (S22)

Teachers noticed the same shift:

“In week two, they wanted the AI to do the work. By week six, they were using it more to test their understanding and to compare different interpretations of the text.” (T1)

This theme reflects a movement from dependence—characterised by uncritical consultation—to more autonomous, self-regulated behaviors in which students attempt to solve comprehension problems independently before using ChatGPT as a confirmatory or elaborative resource. This helps explain the quantitative finding that autonomy increased substantially in the experimental group, even though overall usage frequency was not strongly related to reading improvements.

Theme 3: reconfigured classroom roles and critical awareness of AI

Integrating ChatGPT altered classroom dynamics and triggered discussions about reliability and ethics.

“Because ChatGPT can also be wrong, our teacher always asks, ‘What is your evidence from the text?’ This makes me read more carefully.” (S06)

“Sometimes ChatGPT gave an answer that was not exactly in the text. We checked, and then we talked about why it invented something.” (S19)

“I like that we work in pairs to decide if the AI answer is useful or not. It is not just ‘AI says so.’” (S34)

Teachers framed themselves less as sole providers of information and more as coaches helping students interrogate AI suggestions:

“My role shifted to asking, ‘Do you agree with ChatGPT? Which sentence in the text supports that?’ It opened space for more critical reading.” (T3)

At the same time, concerns emerged about fairness and potential over-use:

“Some students have better phones and internet, so they practise more at home with ChatGPT.” (T4)

This theme, therefore, highlights both the opportunities and tensions that AI introduces: students gain additional sources of explanation and practice, but responsible integration requires explicit attention to critical evaluation, equity of access, and explicit classroom norms. The results are discussed below.

Discussion

The study examined how integrating a ChatGPT-based assistant into B1 CEFR reading instruction influences students’ reading

comprehension and perceived autonomy. Given the clustered classroom design, greater interpretive weight was placed on the mixed-effects models, which explicitly account for class-level variance, while ANCOVA results are provided for transparency.

Quantitatively, linear mixed-effects models indicated that participation in the experimental group was associated with higher posttest reading comprehension and greater perceived reading autonomy after controlling for baseline performance. Qualitatively, these gains were illuminated by students’ and teachers’ accounts of strategic, conditional AI use, rather than frequent or unrestricted reliance on the tool. This integration is particularly evident in the finding that time on task and frequency of AI use were not meaningfully related to learning gains. In contrast, qualitative data showed that how learners engaged with ChatGPT—especially for clarification, inferencing support, and feedback on self-generated summaries—was central to its pedagogical value. This pattern supports process-oriented views of reading development (Xu et al., 2019; Khan and Mutawa, 2021) and aligns with recent calls to move beyond binary evaluations of AI as either beneficial or harmful (Tian and Ironsi, 2025; Al Anqoudi et al., 2023). Rather than functioning as a shortcut, ChatGPT acted as a mediated scaffold that supported comprehension when learners first attempted to engage with the text independently.

From a self-determination theory (SDT) perspective (Mojarrabi, 2024; Zimmerman and Schunk, 2011), the observed gains in perceived autonomy are theoretically coherent. The instructional design emphasized learner choice, responsibility, and self-regulation while constraining AI use through explicit rules and verification requirements. Qualitative findings showed that students increasingly took ownership of reading decisions—such as when to seek help and how to evaluate AI output—suggesting satisfaction of the SDT need for autonomy without undermining competence.

At the same time, the findings resonate with sociocultural theory and the notion of scaffolding (Ariani, 2023; Abdulaal, 2024), in which tools mediate learning processes and are gradually internalized. Classroom observations and learner reflections indicated a developmental shift: early reliance on ChatGPT for clarification gave way to a more selective, reduced use over time. This gradual fading of tool dependence represents a theoretically important contribution, demonstrating that generative AI can function as a temporary scaffold rather than a permanent crutch when pedagogically framed.

Several findings align with emerging empirical work on AI-supported language learning. First, the association between AI-supported instruction and improved comprehension mirrors results reported in recent studies on AI-assisted reading and vocabulary learning (Xu et al., 2019; Istek and Ironsi, 2024; Al Anqoudi et al., 2023). Second, the importance of teacher mediation and explicit guidance aligns with prior research showing that unstructured access to AI yields limited or inconsistent benefits (Ironsi and Ironsi, 2025). Third, learners’ emphasis on verifying AI output against the source text supports arguments for embedding critical AI literacy within language pedagogy (Al Anqoudi et al., 2023; Xu et al., 2019).

However, the study also yields findings that challenge common assumptions in the literature. Notably, contrary to studies suggesting that greater AI engagement leads to greater learning gains (Li and Ironsi, 2025), the present results show no meaningful

relationship between the amount of AI use and reading improvement. Instead, qualitative evidence indicates that excessive or unfocused use was more common among lower-gain students, whereas higher-gain students used ChatGPT sparingly and purposefully. This divergence highlights a key novelty of the study: quality of AI interaction, not quantity, appears to be the critical factor.

Additionally, while some critics argue that AI use inherently undermines learner autonomy and effort (Çelik et al., 2024; Bezirhan and von Davier, 2023), the present findings suggest a more conditional interpretation. When AI use was bounded by pedagogical rules and framed as feedback rather than content generation, learners reported increased confidence and self-regulation rather than dependence. This challenges deterministic narratives about AI and learner passivity and suggests that instructional design mediates AI's motivational effects.

The primary novelty of this study lies in its integrated, classroom-based examination of reading comprehension and autonomy, an area that remains underexplored compared to AI-assisted writing research (Xu et al., 2019). By focusing on reading and by combining mixed-effects quantitative analyses with fine-grained qualitative evidence, the study demonstrates that generative AI can support both cognitive and motivational dimensions of reading when embedded within a structured pedagogy. Moreover, the explicit documentation of developmental change over time—including reduced reliance on AI and increased strategic behavior—adds a temporal dimension often missing from short-term experimental studies. This finding contributes to a more dynamic understanding of AI-supported learning and underscores the importance of longitudinal classroom research.

Despite these contributions, the findings must be interpreted cautiously. The use of intact classes, potential teacher effects, and the limited number of clusters constrain causal inference and generalizability. Accordingly, the results should be understood as evidence that the experimental group was associated with improved outcomes, rather than as proof that ChatGPT alone caused these gains. Nevertheless, the convergence of quantitative and qualitative evidence strengthens confidence in the overall pattern of findings. Taken together, the results suggest that generative AI is neither inherently beneficial nor inherently detrimental to EFL reading development; rather, its educational value depends on pedagogical orchestration, critical use, and alignment with learner autonomy goals.

Conclusion

This study investigated the association between a ChatGPT-integrated reading pedagogy and university EFL learners' reading comprehension and perceived reading autonomy within an authentic classroom context. Using a mixed-methods design and analytic approaches that accounted for classroom clustering, the findings indicate that participation in the experimental group was associated with greater gains in reading comprehension and autonomy than in traditional instruction, while also revealing important nuances about how generative AI operates in reading instruction.

Across quantitative and qualitative strands, the results suggest that ChatGPT's pedagogical value lies not in the amount of use, but in the quality and purposefulness of learner–AI interaction.

Learners who benefited most used ChatGPT selectively—for clarification, inferencing support, and feedback on self-generated summaries—while maintaining primary engagement with the reading text. Qualitative evidence further indicates that structured guidance and explicit rules fostered critical AI literacy, enabling students to question, verify, and gradually reduce reliance on AI support over time. This developmental pattern supports theoretical views of AI as a temporary scaffold that can be internalized rather than a substitute for reading processes.

The study contributes to the growing literature on generative AI in language education by shifting the focus from writing to reading comprehension, and by demonstrating that AI-supported instruction can align with learner autonomy when grounded in principled pedagogical design. The findings also challenge the assumption that increased AI use necessarily leads to better learning outcomes, highlighting the central role of instructional framing and strategic engagement.

At the same time, the conclusions must be interpreted cautiously. The use of intact classes, the small number of clusters, and potential teacher effects limit causal inference and generalizability. Accordingly, the findings should be viewed as preliminary evidence of association rather than definitive proof of effectiveness. This study suggests that generative AI tools such as ChatGPT can play a constructive role in EFL reading instruction when deliberately and critically integrated, in ways that promote learner agency. Future research employing randomized designs, larger numbers of classrooms, and more robust measurement will be essential to clarify further the conditions under which AI-supported reading pedagogy is most effective.

Implications

The findings of this study offer several implications for instructional practice, curriculum design, and future research on generative AI in language education. The findings of this study suggest that ChatGPT can be productively integrated into B1-level reading instruction when it is positioned as a conditional scaffold rather than as a replacement for reading processes. Quantitative and qualitative results converge in showing that learning gains were associated not with the frequency of AI use, but with instructionally regulated, purpose-driven engagement. For classroom practice, this implies that teachers should embed generative AI at clearly defined moments in the reading cycle—after learners have attempted comprehension independently. For example, students may be required to (a) predict word meanings, (b) attempt sentence interpretation, or (c) draft a summary before consulting ChatGPT. AI use can then be limited to clarification, inferencing support, or feedback on learner-generated responses, rather than providing complete explanations or summaries. Such sequencing preserves learners' primary engagement with the text while allowing AI to function as a temporary support.

In addition, there is a need for AI literacy before engaging in classroom reading task. For instance, teachers should explicitly model how to formulate prompts that elicit clarification—not ready-made answers—and gradually shift responsibility to learners to encourage self-regulated AI use. Through this approach, rather than replacing the text, AI can be used to support pre-reading activation such as predicting key vocabulary, generating simplified explanations. Importantly, learners should be required to return to

the original text to verify meaning and identify textual evidence. This design reduces cognitive overload and reading anxiety while preserving depth of processing, inferencing, and guessing skills of students.

The findings also highlight the importance of explicit AI-use rules and teacher mediation. Rather than allowing unrestricted AI access, teachers' explicit guidance—through prompt modeling, rules for acceptable use, and reminders to verify AI output—was integral to learners' engagement with the tool. Accordingly, professional development initiatives should prioritize pedagogical competencies related to AI orchestration, including task design, prompt scaffolding, and strategies for fostering critical evaluation of generative AI output, rather than focusing solely on technical tool training. Teachers in the study guided learners in how to phrase prompts, when AI consultation was appropriate, and how to evaluate AI responses against textual evidence. In practice, this suggests that teachers should model effective prompts, require textual justification of AI output, and gradually reduce guidance to promote internalization of reading strategies and learner autonomy.

At the curriculum and institutional level, the results support treating generative AI as a pedagogical resource that requires intentional integration, rather than solely as a challenge to academic integrity. Curricula can explicitly incorporate AI-supported reading tasks that align with CEFR B1 descriptors, such as identifying main ideas, making simple inferences, and summarizing informational texts, while embedding expectations for verification and reflection. Embedding learning outcomes related to critical AI literacy—such as verifying claims against textual evidence, recognizing AI overgeneralization, and reflecting on ethical and privacy considerations—may help learners navigate AI-supported reading environments responsibly. However, the study also highlights the need to address equitable access, as differences in device or connectivity availability could affect students' opportunities to benefit from AI-supported instruction.

The findings further suggest that critical AI literacy outcomes should be made explicit in reading curricula, including skills such as evaluating the accuracy of AI-generated explanations, recognizing overgeneralization, and reflecting on ethical and privacy considerations. Attention to equitable access is also necessary to ensure that all learners have comparable opportunities to engage meaningfully with AI-supported instruction.

For researchers and educational technologists, the findings suggest that future AI-supported reading tools should prioritize features that encourage self-regulation and text-based reasoning, such as prompting learners to justify interpretations with textual evidence or offering feedback on learner-generated responses rather than providing complete answers. The findings underscore the need to move beyond measuring frequency of AI use and toward examining qualitative dimensions of learner–AI interaction, such as timing, purpose, and strategic intent. From a design perspective, AI tools for reading instruction may be most effective when they prompt learners to justify interpretations with textual evidence or provide feedback on learner-generated responses, rather than offering direct answers.

From a research perspective, the study points to several avenues for further investigation, including replication with randomized or multi-site designs, extension to other CEFR levels and language skills, and longitudinal examination of how AI-supported scaffolding interacts with individual differences in motivation and self-regulation over time. These implications reinforce the conclusion

that the educational review value of generative AI in reading instruction is contingent on pedagogical design, critical framing, and teacher mediation, rather than on the technology itself.

Limitations

A significant limitation of the present study concerns teacher effects and class-level confounding. Because a different instructor taught each intact class and teachers were not crossed or balanced across instructional conditions, it is not possible to disentangle the effects of the instructional condition from potential teacher-related influences. Differences in teaching experience, classroom management, rapport with students, or enthusiasm for the instructional approach may have contributed to the observed group differences.

Although efforts were made to standardize instruction through a shared syllabus, standard texts, common assessments, and before-study briefing, these measures cannot fully control for teacher variability in authentic classroom settings. Consequently, the findings should not be interpreted as evidence that the observed gains were caused solely by the ChatGPT-integrated pedagogy. Instead, the results indicate that participation in a ChatGPT-integrated instructional context, as implemented by the participating teachers, was associated with greater gains in reading comprehension and perceived autonomy.

Future studies should address this limitation by employing designs in which teachers are crossed with conditions, by including teacher as a random effect in studies with larger numbers of classes, or by using randomized or counterbalanced instructional assignments where feasible.

Data availability statement

The original contributions presented in the study are included in the article/[Supplementary material](#), further inquiries can be directed to the corresponding author.

Ethics statement

The studies involving humans were approved by the Akdeniz Karpaz University Ethics Committee. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study. Written informed consent was obtained from the individual(s) for the publication of any potentially identifiable images or data included in this article.

Author contributions

CI: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Writing – original draft, Writing – review & editing. HB: Formal analysis, Methodology, Supervision, Writing – review & editing.

Funding

The author(s) declared that financial support was not received for this work and/or its publication.

Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Generative AI statement

The author(s) declared that Generative AI was not used in the creation of this manuscript.

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Supplementary material

The Supplementary material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/feduc.2026.1763238/full#supplementary-material>

References

- Abdulaal, M. A. A. D. (2024). A sociocultural perspective on the impact of instructor scaffolding on lexical processing and reading comprehension. *SAGE Open* 14, 1–15. doi: 10.1177/21582440241257915
- Al Anqoudi, S., Gabarre, C., Gabarre, S., and Al Aamri, S. M. S. (2023). Improving EFL learners' reading motivation with intensive reading using "ReadTheory". *Int. J. Linguist.* 15, 30–50. doi: 10.5296/ijl.v15i5.21389
- Alfiani, N., et al. (2024). Developing CEFR-based digital literacy resources for multisen-sory English learning. *J. Engl. Teach. Learn. Educ.* 17, 49–73. doi: 10.24042/ee-jtbi.v17i1.20934
- Al-Hoorie, A. H., Oga-Baldwin, W. Q., Hiver, P., and Vitta, J. P. (2022). Self-determination mini-theories in second language learning: a systematic review of three decades of research. *Lang. Teach. Res.* 22, 1–29.
- Alrabai, F. (2020). Enhancing reading skill via ReadTheory.Org: students' attitudes, motivation, autonomy, and perceptions. *Arab World Engl. J.* 11, 1–81. doi: 10.24093/awej/th.273
- Ariani, D. (2023). The effectiveness of interactive reading models in English reading instruction. *Int. J. Learn. Teach. Educ. Res.* 22, 124–141.
- Azhar, A., and Abdullah, A. (2024a). Artificial intelligence (AI) in language learning autonomy (LLA) – a systematic literature review uncovering learning autonomy. *Int. J. Res. Found. Multidiscip. Res.* 6:31045. doi: 10.36948/ijfmr.2024.v06i06.31045
- Azhar, A., and Abdullah, A. (2024b). Artificial intelligence (AI) in language learning autonomy (LLA)—a systematic literature review uncovering learning autonomy. *SSRN Electron. J.*, 1–30. doi: 10.2139/ssrn.5658651
- Bezirhan, U., and von Davier, M. (2023). Automated reading passage generation with OpenAI's large language model. *Comput. Educ. Artif. Intell.* 5:100161. doi: 10.1016/j.caeai.2023.100161
- Braun, V., and Clarke, V. (2022). Toward good practice in thematic analysis: avoiding common problems and maximizing benefits. *Asia Pac. J. Qual. Res.* 24, 1–6. doi: 10.1080/26895269.2022.2129597
- Byrne, D. A. (2022). Worked example of Braun and Clarke's approach to reflexive thematic analysis. *Qual. Quant.* 56, 1391–1412. doi: 10.1007/s11135-021-01182-y
- Çelik, F., Yangin-Ersanlı, C., and Arslanbay, G. (2024). Does AI simplification of authentic blog texts improve reading comprehension, inferencing, and anxiety? A one-shot intervention in the Turkish EFL context. *Int. Rev. Res. Open Distrib. Learn.* 25, 287–303. doi: 10.19173/irrodl.v25i3.7779
- Chen, X., Hu, Z., and Wang, C. (2024). Empowering education development through AIGC: a systematic literature review. *Educ. Inf. Technol.* 29, 17485–17537. doi: 10.1007/s10639-024-12549-7
- Council of Europe (2020). *Common European framework of reference for languages: Learning, teaching, assessment – Companion volume*. Strasbourg: Council of Europe.
- Creswell, J. W., and Plano Clark, V. L. (2018). *Designing and conducting mixed methods research*. 3rd ed. Thousand Oaks: SAGE.
- Dehghan, F. (2013). Sociocultural theory and reading comprehension: the scaffolding of readers in an EFL context. *Int. J. Res. Stud. Lang. Learn.* 2, 67–80. doi: 10.5861/ijrsl.2012.183
- Dempere, J., Modugu, K., Hesham, A., and Ramasamy, L. K. (2023). The impact of ChatGPT on higher education. *Front. Educ.* 8:1206936. doi: 10.3389/feduc.2023.1206936
- Fetters, M. D., Curry, L. A., and Creswell, J. W. (2013). Achieving integration in mixed methods designs—principles and practices. *Health Serv. Res.* 48, 2134–2156. doi: 10.1111/1475-6773.12117
- Field, A. (2018). *Discovering statistics using IBM SPSS statistics*. 5th Edn. London: SAGE.
- Hidayat, M. T. (2024). Effectiveness of AI-based personalised reading platforms in enhancing comprehension skills. *J. Learn. Dev.* 11, 115–125. doi: 10.56059/jl4d.v11i1.955
- Irons, C. S. (2024). "Exploring the potential of generative AI in English language teaching" in *Facilitating global collaboration and knowledge sharing in higher education with generative AI*. eds. P. Yu, J. Mulli, Z. A. S. Syed and L. Umme (Hershey, PA, USA: IGI Global Scientific Publishing), 162–185.
- Irons, C. S., and Irons, S. S. (2025). Experimental evidence for the efficacy of generative AI in improving students' writing skills. *Qual. Assur. Educ.* 33, 237–252. doi: 10.1108/QAE-04-2024-0065
- Istek, E., and Irons, C. S. (2024). Incorporating mobile learning and the CAP(E) framework in improving the receptive skills of students. *Libr. Hi Tech* 42, 711–729. doi: 10.1108/LHT-01-2023-0008
- Jiang, L. (2022). How does artificial intelligence empower EFL teaching and learning nowadays? *Front. Psychol.* 13:1049401. doi: 10.3389/fpsyg.2022.1049401
- Khan, A., and Mutawa, M. (2021). Enhancing reading comprehension skills of Arab EFL learners using an AI-based personalised reading platform. *Int. J. Emerg. Technol. Learn.* 16, 119–136.
- Kim, Y.-S. G. (2020). Interactive and dynamic literacy model: an integrative theoretical framework for reading development. *Educ. Psychol. Rev.* 32, 1–31.
- Korkut, N. (2023). AI-enhanced English language teaching in K–12 education. *Educ. Sci.* 13, 1–14.
- Li, Z., and Irons, C. S. (2025). The efficacy of generative artificial intelligence in developing science education preservice teachers' writing skills: an experimental approach. *J. Sci. Educ. Technol.* 34, 1260–1271. doi: 10.1007/s10956-024-10148-2
- Lo, C. K., Yu, P. L. H., Xu, S., Ng, D. T. K., and Jong, M. S. Y. (2024). Exploring the application of ChatGPT in ESL/EFL education and related research issues: a systematic review of empirical studies. *Smart Learn. Environ.* 11:50. doi: 10.1186/s40561-024-00342-5

- Ma, Y., et al. (2025). The human touch in AI: optimizing language learning through need-supportive design. *Front. Psychol.* 16:1568239.
- Macaskill, A., and Taylor, E. (2010). The development of a brief measure of learner autonomy in university students. *Studies in higher education*, 35, 351–359.
- McEown, M. S. (2019). Self-determination for all language learners. *System* 86, 102124–102124. doi: 10.1016/j.system.2019.102124
- Mojarrabi, H. (2024). The comparative effects of teacher versus peer-scaffolding on EFL learners' incidental vocabulary learning and reading comprehension. *J. Teach. Lang. Skills* 43, 89–113.
- Mynard, J. (2022). Self-determination theory: a proposed framework for self-access language learning. *J. Pers. Learn. Learn. Environ.* 2, 1–20. doi: 10.21832/9781788929059-003
- Oğuz, A. (2013). Developing a Scale for Learner Autonomy Support. *Educational Sciences: Theory and Practice*, 13, 2187–2194.
- Spratt, M., Humphreys, G., and Chan, V. (2002). Autonomy and motivation: Which comes first? *Language teaching research* 6, 245–266.
- Tabachnick, B. G., and Fidell, L. S. (2019). *Using multivariate statistics*. 7th Edn. Boston: Pearson.
- Tian, X., and Ironsi, C. S. (2025). Examining the impact of augmented reality on students' learning outcomes. *Sci. Rep.* 15:36957. doi: 10.1038/s41598-025-20833-w
- Tram, N. H. M., Nguyen, T. T., and Tran, C. D. (2024). ChatGPT as a tool for self-learning English among EFL learners: a multi-methods study. *System* 127:103528. doi: 10.1016/j.system.2024.103528
- Wang, S., Wang, F., Zhu, Z., Wang, J., Tran, T., and Du, Z. (2024). Artificial intelligence in education: a systematic literature review. *Expert Syst. Appl.* 252:124167. doi: 10.1016/j.eswa.2024.124167
- Wei, L. (2023). Artificial intelligence in language instruction: impact on English learning achievement, L2 motivation, and self-regulated learning. *Front. Psychol.* 14:1261955. doi: 10.3389/fpsyg.2023.1261955
- Xiao, C., Xu, SX, Zhang, K, Wang, Y, and Xia, L (2023). "Evaluating reading comprehension exercises generated by LLMs: a showcase of ChatGPT in education applications," in *Proceedings of the 18th Workshop on Innovative Use of NLP for Building Educational Applications*. Toronto: Association for Computational Linguistics, 610–625.
- Xu, Z., Wijekumar, K., Ramirez, G., Hu, X., and Irej, R. (2019). The effectiveness of intelligent tutoring systems on K–12 students' reading comprehension: a meta-analysis. *Br. J. Educ. Technol.* 50, 3119–3137. doi: 10.1111/bjet.12758
- Yawiloeng, R. (2021). Peer scaffolding during EFL reading activities: a sociocultural perspective. *TESOL J.* 14:e00569. doi: 10.5539/elt.v14n12p44
- Yousefi, M., and Askari, M. I. (2024). Exploring the effectiveness of artificial intelligence (AI) on reading comprehension among Iranian EFL learners. *J New Trends Educ Lang Learn* 5, 1–15.
- Zhang, Y., and Liu, H. (2024). Artificial intelligence in language learning: a systematic review. *Front. Psychol.* 15:1402567.
- Zimmerman, B. J. (2002). Becoming a self-regulated learner: an overview. *Theory Pract.* 41, 64–70. doi: 10.1207/s15430421tip4102_2
- Zimmerman, B. J., and Schunk, D. H. (2011). *Handbook of self-regulation of learning and performance*. New York, NY: Routledge.